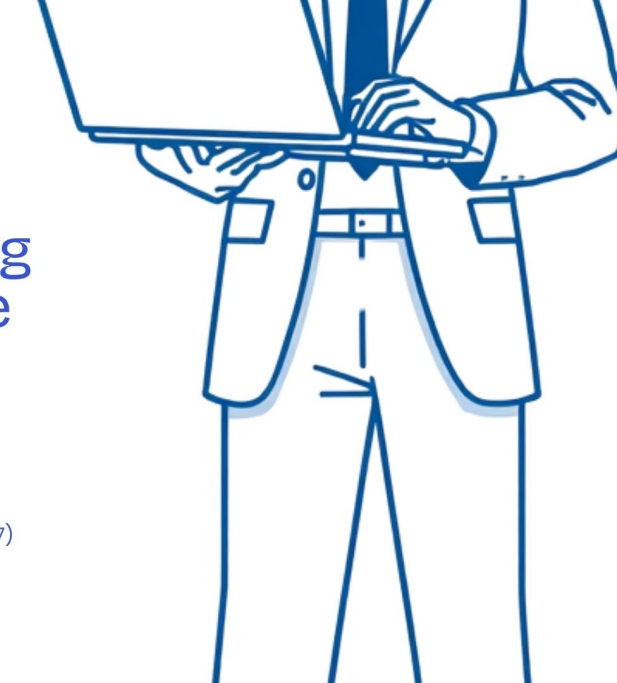
SECP
3133
Section 2

# Optimizing High-Performance Data Processing For Large-Scale Web Crawlers

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# Project Background

**Problem:**

- News websites publish thousands of articles daily
- Manual data collection is slow and inefficient
- Large datasets require efficient crawling and processing

**Proposed Solution:**

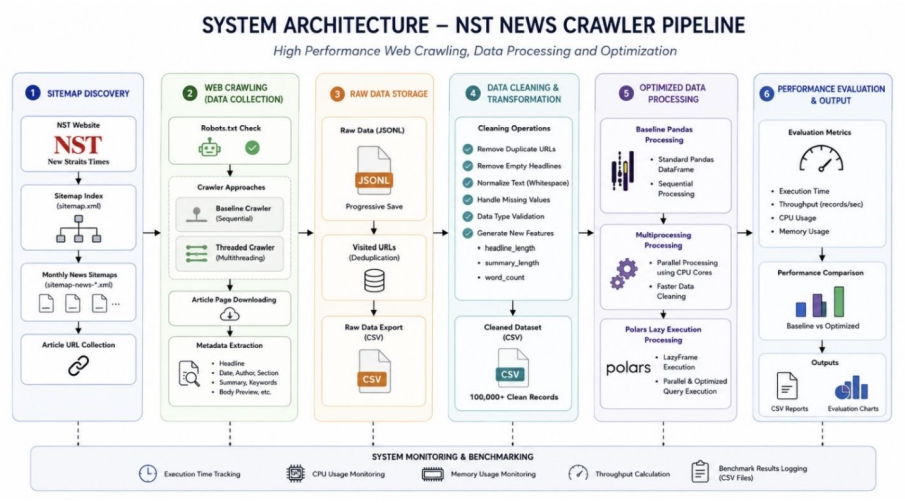
Develop a **high-performance NST web crawler** capable of collecting over 100,000 article metadata records with optimized data processing

NEW STRAITS TIMES

# Objectives

- Develop an automated web crawler for NST
- Collect more than 100,000 news records
- Store data in JSONL and CSV formats
- Improve performance using:
  - Multithreading
  - Multiprocessing
  - Polars Lazy Execution
- Evaluate execution time, CPU usage, memory usage and throughput

# System Architecture



# Data Collection

01

**TARGET WEBSITE**

New Straits Times (NST)

02

**METADATA COLLECTED**

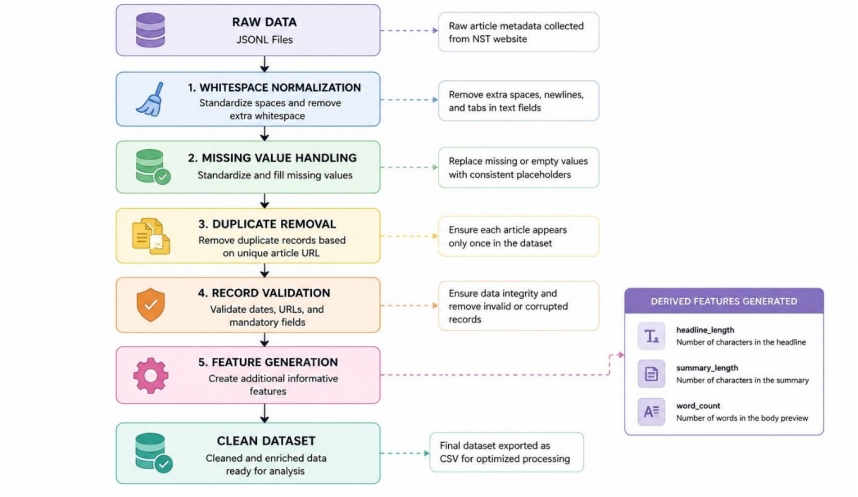
- Headline
- Article URL
- Publication Date
- Modified Date
- Author
- Section
- Summary
- Keywords
- Body Preview
- Word Count
- Collection Timestamp

03

**RESULT**

✅ Successfully collected 102,265 news article records

# Data Processing



# Optimization Techniques

01

### MULTITHREADED WEB CRAWLING

- ThreadPoolExecutor
- Crawl multiple pages simultaneously
- Faster downloading

02

### PARALLEL DATA CLEANING

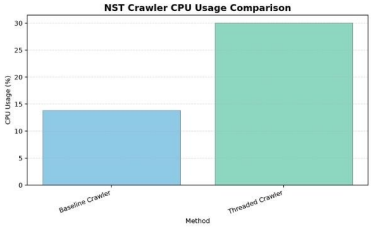
- Parallel data cleaning
- Multiple CPU cores
- Faster preprocessing

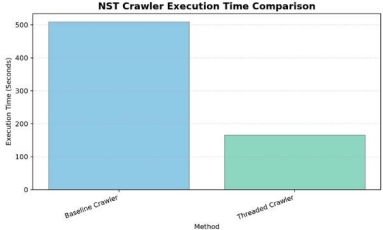
03

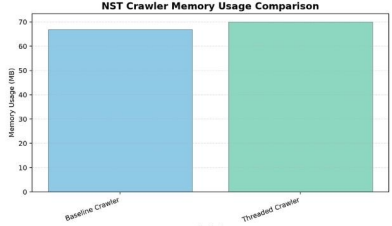
### POLARS LAZY EXECUTION

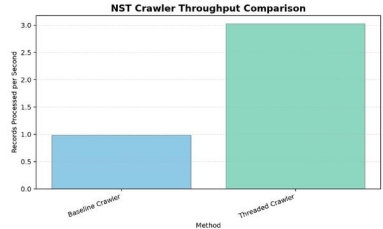
- LazyFrame
- Query optimization
- Automatic parallel execution

# Performance Evaluation - Web Crawling



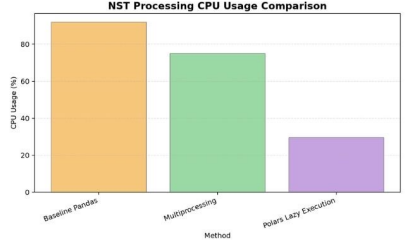



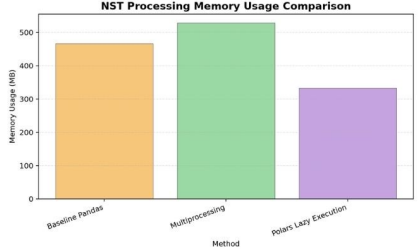

**Performance Summary**

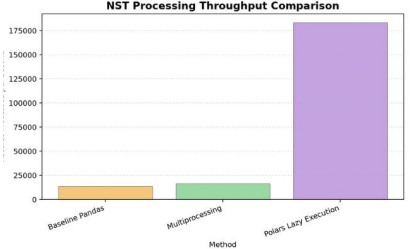
- Multithreading significantly reduced crawling execution time.
- Throughput increased significantly while maintaining data quality.
- System resources were utilized more efficiently than the baseline crawler.

# Performance Evaluation - Data Processing








**Performance Summary**

- Multiprocessing improved the efficiency of data cleaning.
- Polars Lazy Execution achieved the shortest execution time and the highest throughput.
- Polars provided the best overall processing performance with lower CPU and memory usage than the baseline.

# Challenges & Limitations

**Challenges Encountered**

🌐 Network Dependency
- Crawling performance depends on internet speed and NST server response time.
- Execution time may vary between benchmark runs.

⚖️ Balancing Speed and Ethics
- Too many threads can overload the target website.
- Number of threads and request delays were carefully configured.

🖥️ Multiprocessing Overhead
- Creating and managing multiple processes consumes additional system resources.
- Process synchronization is required before merging results.

**Limitations**

📄 Website-Specific Evaluation
- Performance results are based only on the NST website.
- Results may differ for websites with different structures or dynamic content.

📊 Workload Dependency
- Polars performs best for the current dataset.
- Different datasets or analytical tasks may produce different results.

💻 Hardware Dependency
- Performance depends on CPU, memory, operating system, and network conditions.
- Results may vary across different computing environments.

# Conclusion & Future Work

**Concusion**

✔️ Built an automated NST web crawler
✔️ Collected 102,265 records
✔️ Successfully implemented
- Multithreading
- Multiprocessing
- Polars Lazy Execution

✔️ Performance improved significantly

**Future Work**

🚀 Asynchronous Crawling
- AsyncIO / Async HTTP libraries

🌐 Multiple News Sources
- Support multiple Malaysian news website

⚡ Distributed Processing
- Apache Spark or Dask
- Handle millions of records

🤖 Advanced News Analytics
- Topic Classification
- Sentiment Analysis
- Keyword Extraction
- Trend Analysis


Thank
you